

2016.0 RANGE ROVER (LG), 415-00

INFORMATION AND ENTERTAINMENT SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the information and entertainment system, refer to the relevant description and operation sections in the workshop manual.

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- All diagnostic equipment should comply with local legislation.
- Relevant diagnostic equipment should be regularly checked and calibrated according to the manufacturer's instructions.
- The workshop should be equipped with a full range of general equipment which is to be kept in good order and available to all suitably trained staff.
- Diagnostic equipment must meet the JLR Minimum Standards for general equipment as outlined in TOPIx.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Integrated Audio Module (IAM) ▪ Audio Amplifier Module (AAM) ▪ Touch Screen (TS)	▪ Fuses ▪ Wiring harnesses and connectors ▪ Integrated Audio Module (IAM)

▪ Satellite Radio Control Module (SRCM) ▪ Digital Radio Control Module (DRCM) ▪ TV Control Module (TVCM) ▪ Rear Seat Entertainment Control Module (RSECM) ▪ Rear View Camera (RVC) ▪ Speakers ▪ Scratched/dirty compact discs ▪ Water ingress	▪ Audio Amplifier Module (AAM) ▪ Touch Screen (TS) ▪ Satellite Radio Control Module (SRCM) ▪ Digital Radio Control Module (DRCM) ▪ TV Control Module (TVCM) ▪ Rear Seat Entertainment Control Module (RSECM) ▪ Rear View Camera (RVC) ▪ Antennae ▪ Speakers
--	---

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Audio/video system inoperative at start up	▪ Last used audio/video source inoperative ▪ MOST network fault	▪ GO to Pinpoint Test A.
Audio/video soft key greyed out on touch screen	▪ Last used audio/video source inoperative ▪ MOST network fault	▪ GO to Pinpoint Test A.
Soft key response different to soft key	▪ Touch screen calibration incorrect	▪ Perform touch screen calibration. GO to Pinpoint Test C.

touched		
Shortcut soft keys missing from home screen on the touch screen	▪ Shortcut soft keys set to off after operating the Clear all soft key on the 'Home menu shortcuts' screen	▪ Set the shortcut soft keys to on by navigating to the home screen and operating the following soft keys: Setup, System, Home menu shortcuts, Default. The default shortcut soft keys will now be available on the home screen. The shortcut soft keys can be customised as necessary
Poor audio quality (all sources)	▪ MOST harness connections loose ▪ MOST harness connections contaminated ▪ MOST harness misrouted - Too many bends or bend radius less than 25mm ▪ Audio amplifier system fault	▪ Check MOST harness connectors for security ▪ Check MOST harness connectors for contamination ▪ Check the routing of the MOST harness ▪ Using the manufacturer approved diagnostic system, check the audio amplifier module for related DTCs and refer to the relevant DTC index
One or more speaker(s) inoperative	▪ Audio amplifier system fault ▪ Speaker system fault	▪ Using the manufacturer approved diagnostic system, check the audio amplifier module for related DTCs and refer to the relevant DTC index ▪ For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual REFER to: Speakers (415-01 Information and Entertainment System, Diagnosis and Testing).
One or more speaker(s) poor or weak sound	▪ Speaker system fault	▪ For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual
One or more speaker(s) vibrating (Buzzing) excessively	▪ Speaker system fault	▪ For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual
Sub-woofer inoperative	▪ Sub-woofer system	▪ For Speaker Diagnostics please refer

	fault	to section 415-01 Diagnosis and Testing - Speakers in the workshop manual
Sub-woofer poor or weak sound	Sub-woofer system fault	For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual
Sub-woofer vibrating (Buzzing) excessively	Sub-woofer system fault	For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual
Sub-woofer output distorted	Sub-woofer system fault	For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual
Radio inoperative	- AM/FM antenna fault - MOST network fault - Power supply failure - Integrated audio module internal failure	- Check if other audio sources activate the speakers - Check the harness for signs of damage - Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
Digital radio inoperative	- Digital radio antenna fault - Tuner failure - Antenna connectivity or harness - MOST network fault - Power supply failure - Digital radio control module internal failure	- Check the harness for signs of damage - Using the manufacturer approved diagnostic system, check the digital radio control module for related DTCs and refer to the relevant DTC index
Digital radio no signal reception - New digital radio control module	- Initial tuning not completed - Tuner failure - Defective component	

	in antenna circuit - Antenna signal reception is obstructed by buildings, clouds, trees or tunnels - Low MOST signal - Extreme temperature in vehicle - Water ingress	NOTE: Some functions are inhibited when the vehicle is moving. Operate the Auto-tune soft key
Digital radio poor signal reception	- Tuning not refreshed Link DAB set to off - L-band antenna set to off in a L-band antenna region - Band 3 antenna set to off in a band 3 antenna region - L-band antenna or link harness damaged - Band 3 antenna or link harness damaged - Software issue - Antenna damaged - Antenna signal reception is obstructed by buildings, clouds and trees or tunnels - Low MOST signal - Extreme temperature in vehicle - Water ingress	NOTE: Some functions are inhibited when the vehicle is moving. - Operate the Auto-tune soft key - Operate the Settings soft key, followed by the Options soft key and set Link DAB to on - Operate the Settings soft key, followed by the L band and Band 3 soft key - Operate the Settings soft key, followed by the L band and Band 3 soft key - Check the L-band antenna and link harness for damage - Check the band 3 antenna and link harness for damage - Using the manufacturer approved diagnostic system, re-configure the digital radio control module with the latest level software - Check the antenna for damage
Digital radio channel list not displayed	- Tuning not refreshed - L-band antenna set to off in a L-band antenna region - Band 3 antenna set to off in a band 3 antenna region	NOTE: Some functions are inhibited when the vehicle is moving.

	▪ MOST connector damaged ▪ MOST network issue	▪ Operate the Auto-tune soft key ▪ Operate the Settings soft key, followed by the L band and Band 3 soft key ▪ Operate the Settings soft key, followed by the L band and Band 3 soft key ▪ Check MOST harness connectors for damage
Digital radio interrupted by announcements	▪ Announcements set to on ▪ MOST connector damaged ▪ MOST network issue ▪ Moving through a tunnel	NOTE: Some functions are inhibited when the vehicle is moving. ▪ Operate the Settings soft key, followed by the Announcements soft key, and set all Announcements to off ▪ Check MOST harness connectors for damage
Unable to store preset channels in digital radio	▪ Preset # soft key not operated for sufficient duration	▪ Operate the Preset # soft key for at least 2 seconds to store the current station
Digital radio will not select preset station when Preset # soft key operated	▪ No station stored to relevant Preset # soft key	▪ Store a station to the relevant Preset # soft key and retest
NOTE: Satellite Digital Audio Radio Service (SDARS) applies to NAS market vehicles only. Satellite radio inoperative	▪ Satellite radio antenna fault ▪ Satellite radio control module internal failure	▪ Using the manufacturer approved diagnostic system, check the satellite radio control module for related DTCs and refer to the relevant DTC index

NOTE: Satellite Digital Audio Radio Service (SDARS) applies to NAS market vehicles only. No Satellite Digital Audio Radio Service (SDARS) channels available - 'No signal' or 'Acquiring signal' message	▪ Vehicle not in USA ▪ Poor signal reception ▪ Satellite Digital Audio Radio Service (SDARS) system fault	▪ No fault to rectify ▪ The vehicle must have clear line of sight to a satellite, which means being outside and having no blockages between the vehicle and the satellite ▪ Using the manufacturer approved diagnostic system, check the Satellite Radio Control Module (SRCM) for related DTCs and refer to the relevant DTC index
NOTE: Satellite Digital Audio Radio Service (SDARS) applies to NAS market vehicles only. Only one Satellite Digital Audio Radio Service (SDARS) channel available (channel 184) - 'Unsubscribed' message	▪ Channel 184 is available without subscription for a limited period (6 months)	▪ No fault to rectify. Subscribe to Satellite Digital Audio Radio Service (SDARS) to resume service
NOTE: Satellite Digital Audio Radio Service (SDARS) applies to NAS market vehicles only.	▪ Channel 184 is available without subscription for a limited period (6 months)	▪ No fault to rectify. Subscribe to Satellite Digital Audio Radio Service (SDARS) to resume service

No Satellite Digital Audio Radio Service (SDARS) channels available after 6 months without subscribing - 'Unsubscribed' message		
NOTE: Satellite Digital Audio Radio Service (SDARS) applies to NAS market vehicles only. Some Satellite Digital Audio Radio Service (SDARS) unavailable - 'Unsubscribed' message	▪ Adult channels blocked by subscription type (family package)	▪ No fault to rectify. Subscribe to Satellite Digital Audio Radio Service (SDARS) full package to receive adult channels
NOTE: Satellite Digital Audio Radio Service (SDARS) applies to NAS market vehicles only. Satellite Digital Audio Radio Service (SDARS) pay channels unavailable - 'Unsubscribed' message	▪ Payment not made	▪ No fault to rectify
Compact disc player inoperative	▪ Incompatible/damaged compact disc ▪ Integrated audio module internal failure	▪ Insert a known good disc and retest ▪ Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC

		index
Unable to upload files to the hard drive	▪ Incompatible/damaged compact disc ▪ Integrated audio module internal failure	▪ Insert a known good disc and retest ▪ Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
Auxiliary audio inoperative	▪ Incompatible/faulty auxiliary device ▪ Auxiliary device link cable fault ▪ Integrated audio module internal failure	▪ Connect a known good auxiliary device to the auxiliary socket and retest ▪ Connect a known good auxiliary device to the auxiliary socket using a known good link cable and retest ▪ Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
USB audio/video inoperative	▪ Incompatible/faulty USB device ▪ Integrated audio module internal failure	▪ Connect a known good USB device to the auxiliary socket and retest ▪ Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
USB audio/video inoperative - Apple devices	▪ Incompatible/faulty Apple device ▪ Bluetooth® and USB connections made in the incorrect order ▪ Integrated audio module internal failure	▪ Check Apple device compatibility table below. Connect a known good Apple device to the auxiliary socket and retest ▪ Audio streaming is supported via the USB cable but this must be connected after the cellular phone connects via Bluetooth® - Best practice is to start the engine (causing the Bluetooth® connection to be made) before connecting the USB cable ▪ Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index

Television inoperative	- TV antenna fault - TV control module internal failure	NOTE: Some functions are inhibited when the vehicle is moving. Using the manufacturer approved diagnostic system, check the TV control module for related DTCs and refer to the relevant DTC index
Television video signal poor/inoperative at the touch screen (television audio normal) - Vehicles without rear seat entertainment	CVBS signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTES: - Some functions are inhibited when the vehicle is moving. - The television audio signal is transmitted on the MOST network. - Using the manufacturer approved diagnostic system, check the television control module for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the CVBS signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the television control module for related DTCs and refer to the relevant DTC index
Television video signal poor/inoperative at the touch screen (television audio normal) - Vehicles with rear seat entertainment	- Antenna damaged - Antenna circuit short circuit to ground, short circuit to power, open circuit, high resistance - LVDS signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTES: - Some functions are inhibited when the vehicle is moving. - The television audio signal is transmitted on the MOST network. - Check the antenna for damage

	▪ CVBS signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Non-genuine electronic accessories	▪ Refer to the electrical circuit diagrams and check the antenna circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the rear seat entertainment control module for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the LVDS signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the rear seat entertainment control module for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the CVBS signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Disconnect/remove non-genuine electronic accessories and retest
Television video signal poor/inoperative at the touch screen AND the rear seat entertainment screens (television audio normal) - Vehicles with rear seat entertainment	▪ Antenna damaged ▪ Antenna circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ LVDS signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ TV control module fault ▪ Non-genuine electronic accessories	NOTE: The television audio signal is transmitted on the MOST network. ▪ Check the antenna for damage ▪ Refer to the electrical circuit diagrams and check the antenna circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the rear seat entertainment control module for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the LVDS signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved

		diagnostic system, check the television control module for related DTCs and refer to the relevant DTC index ▪ Disconnect/remove non-genuine electronic accessories and retest
Television channel list absent	▪ Incorrect country setting ▪ Software fault	NOTE: Some functions are inhibited when the vehicle is moving. ▪ Check country setting and reset as necessary ▪ Set the country setting to Ukraine and wait 60 seconds. If the channel list is now present, reset to the correct country and using the manufacturer approved diagnostic system, re-configure the television control module with the latest level software
Unable to store television preset channels	▪ Preset # soft key not operated for sufficient duration	▪ Operate the Preset # soft key for at least 2 seconds to store the current channel
Television will not select preset channel when Preset # soft key operated	▪ No television channel stored to relevant Preset # soft key	NOTE: Some functions are inhibited when the vehicle is moving. ▪ Store a channel to the relevant Preset # soft key and retest
DVD player inoperative	▪ Incompatible/damaged compact disc ▪ Incorrect region set ▪ Integrated audio module internal failure	NOTE: Some functions are inhibited when the vehicle is moving. ▪ Insert a known good disc and retest ▪ Change region setting

		Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
Unable to pair mobile phone to vehicle via Bluetooth®	Incompatible mobile phone	NOTE: Installing new components will not improve connectivity with an incompatible mobile phone. Check mobile phone compatibility by referring to: www.landrover.com/Owners/Bluetooth and following the instructions on the page
Echo when using a mobile phone via Bluetooth®	Noise cancelling set to On in mobile phone and vehicle	Set noise cancelling to Off in mobile phone
Navigation system inoperative (integrated navigation system)	- Navigation antenna fault - Integrated audio module internal failure	Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
Navigation system inoperative (with navigation control module)	- Navigation antenna fault - Navigation control module internal failure	Using the manufacturer approved diagnostic system, check the navigation control module for related DTCs and refer to the relevant DTC index
Traffic message channel inoperative	- FM/TMC antenna fault - VICS antenna fault - Integrated audio module internal failure	NOTE: Vehicle Information and Communication System (VICS) is a type of Traffic Message Channel system used in the Japan market only. Using the manufacturer approved

		diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
Rear seat entertainment system inoperative	▪ Rear seat entertainment control module internal failure	▪ Using the manufacturer approved diagnostic system, check the rear seat entertainment control module for related DTCs and refer to the relevant DTC index
Poor/no image displayed from rear view camera	▪ Touch screen fault ▪ Rear view camera fault	▪ Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the relevant DTC index ▪ For further camera diagnostics please refer to section 413-13 Diagnosis and Testing - Proximity Camera in the workshop manual. REFER to: Proximity Camera (413-13 Parking Aid, Diagnosis and Testing).

USB/CD DATA DISK AUDIO FILE COMPATIBILITY

NOTE:

Before attempting a repair to the in-vehicle infotainment system following concerns regarding no playback of audio files stored either on a USB data storage device or on a CD data disc, check below to ensure that the audio files in question are encoded in a compatible format

There are a number of variables that can be set (either automatically or by the user) at the point of creating the audio file that may contribute to an audio file being encoded in an incompatible format. These include:

- The **type of audio file** created (MP3/WMA/AAC)

- The specification of **Variable Bit Rate** (VBR) or **Constant Bit Rate** (CBR) encoding
- If CBR encoding is being used, then a particular **bit rate** value (measured in kilobits per second - kbps) may be selected
- The rate of **sampling frequency**, measured in kilohertz (kHz), may also be selected

Diagnostic Procedures For Audio Files

Identify File Type: if a customer reports issues with audio file playback, first confirm that the data source is operating normally and is not locked or corrupted. This may be achieved by reading the USB storage device or data disk via a PC and confirming that the audio files can be seen/accessed as expected. If the storage device/data disk appears to be operating normally, the next step is to ascertain the file type of those files that will not play through the infotainment system. There are three types of compatible audio file, either **MP3** (which must have a file extension of .mp3); or **WMA** (which must have a file extension of .wma); or **AAC** (which must have a file extension of either .aac or .m4a). If the affected audio files are not of these types, then successful playback via the infotainment system may not be possible.

Further information about the audio file may be accessed via the file properties tab either when viewing the file in Windows Explorer or when playing the file via a digital media player programme. In such a way, it should be possible to ascertain some or all of the required information concerning the file's specified encoding type/bit rate/sampling frequency. Once this information has been obtained, use the tables detailed below to check if the suspect audio file is compatible with the vehicle infotainment system.

Playback Of Audio Files Stored On A USB Storage Device

USB MP3 Files (only if file extension is '.mp3'): Playback of MP3 audio files encoded in Variable Bit Rate (VBR) format is supported at bit rates between

8-320 kbps. For Constant Bit Rate (CBR) files, see table below for compatible sampling rates and bit rates.

FILE FORMAT/ENCODING FORMAT	SAMPLING RATE	BIT RATES	NOTES
MP3 (MPEG 2.5*)	All available are supported	8-160 kbps playback supported	*For MPEG 2.5 format audio files, playback cannot be guaranteed but an attempt will be made to play
MP3 (MPEG 2)	All available are supported	8-160 kbps playback supported	-
MP3 (MPEG 1)	All available are supported	32-128 kbps; 160-320 kbps playback supported	Playback of MPEG 1 audio files with a bit rate of 144 kbps is not supported

USB WMA Files (only if file extension is '.wma'): Playback of WMA audio files encoded in Variable Bit Rate (VBR) format and created using Windows Media Player Version 8.0 and/or Version 9.0 is supported at bit rates between 5-384 kbps. For Constant Bit Rate (CBR) files, see table below for compatible sampling rates and bit rates. Note that WMA CBR files created using Windows Media Player Versions 7.0, 8.0 and 9.0 can be supported, while playback will be attempted but cannot be guaranteed for files created using Windows Media Player Versions 4.0, 4.1, 9 Beta and 9.1.

FILE FORMAT/ENCODING FORMAT	SAMPLING RATE	BIT RATES	NOTES
WMA - Constant Bit Rate (CBR)	At sampling rates of 8 KHz playback supported only at specified bit rates	5-8 kbps (mono); 12 kbps (stereo) playback supported. 31 kbps (mono) playback cannot be guaranteed but an attempt will be made to play	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
	At sampling	8-10 kbps (mono) playback	'Lossless', 'Professional' or 'Voice' format files created

rates of 11.03 KHz playback supported only at specified bit rates	supported	in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
At sampling rates of 16 KHz playback supported only at specified bit rates	10-12 kbps and 16 kbps (mono); 16-20 kbps (stereo) playback supported	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
At sampling rates of 22.05 KHz playback supported only at specified bit rates	16-20 kbps (mono); 20, 22 and 36 kbps (stereo) playback supported	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
At sampling rates of 32 KHz playback supported only at specified bit rates	20 kbps (mono); 32, 40, 48 kbps (stereo) playback supported. 32 kbps (mono); 22, 36, 44, 64, 384 kbps (stereo) playback cannot be guaranteed but an attempt will be made to play	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
At sampling rates of 44.1 KHz playback supported only at specified bit rates	20, 32 and 48 kbps (mono); 32, 48, 64, 80, 96, 128, 160, 192, 256 and 320 kbps (stereo) playback supported. 15 kbps (mono) playback cannot be guaranteed but an attempt will be made to play	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
At	64, 96, 128, 160,	'Lossless', 'Professional' or

	sampling rates of 48 KHz playback supported only at specified bit rates	192 and 256* kbps (stereo) playback supported. 32 kbps (mono) & 48, 63, 95, 127, 191 and 320 kbps (stereo) playback cannot be guaranteed but an attempt will be made to play	'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported. *All available versions can be supported at a bit rate of 256 kbps for this sampling rate only
--	---	--	---

USB AAC Files (only if file extension is '.aac' or '.m4a'): Playback of AAC audio files encoded in Variable Bit Rate (VBR) format is supported at bit rates between 8-320 kbps. For Constant Bit Rate (CBR) files, see table below for compatible sampling rates and bit rates.

FILE FORMAT/ENCODING FORMAT	SAMPLING RATE	BIT RATES	NOTES
AAC - Constant Bit Rate (CBR)	At sampling rates of between 8-32 KHz, playback cannot be not guaranteed but may be possible at some bit rates	8-320 kbps playback cannot be not guaranteed, but may be possible in some cases	-
	At sampling rates of between 44.1 - 48 KHz, playback supported at specified bit rates	48-80 kbps; 96-128 kbps; 160-256 kbps; 320 kbps playback supported	Playback at other bit rates between 44.1-576 kbps may be possible, but cannot be verified
	At sampling rates of between 64-96 KHz, playback cannot be not guaranteed but an attempt will be made to play at some bit rates	96-768 kbps playback cannot be not guaranteed but an attempt will be made to play	Playback at other bit rates between 64-1152 kbps cannot be not guaranteed but an attempt will be made to play

Playback Of Audio Files Stored On A CD Data Disk

CD Data Disk MP3 Files (only if file extension is '.mp3'): Playback of MP3

audio files encoded in Variable Bit Rate (VBR) format is supported at bit rates between 8-320 kbps. For Constant Bit Rate (CBR) files, see table below for compatible sampling rates and bit rates.

FILE FORMAT/ENCODING FORMAT	SAMPLING RATE	BIT RATES	NOTES
MP3 (MPEG 2.5*)	All available are supported	8-160 kbps playback supported	*For MPEG 2.5 format audio files, playback cannot be guaranteed but an attempt will be made to play
MP3 (MPEG 2)	All available are supported	8-160 kbps playback supported	-
MP3 (MPEG 1)	All available are supported	32-128 kbps; 160-320 kbps playback supported	Playback of MPEG 1 audio files with a bit rate of 144 kbps is not supported

CD Data Disk WMA Files (only if file extension is '.wma'): Playback of WMA audio files encoded in Variable Bit Rate (VBR) format and created using Windows Media Player Version 9.0 is supported at bit rates between 32-192 kbps. For Constant Bit Rate (CBR) files, see table below for compatible sampling rates and bit rates. Note that WMA CBR files created using Windows Media Player Version 9.0 can be supported, while playback will be attempted but cannot be guaranteed for files created using Windows Media Player Versions 4.0, 4.1, 7.0, 8.0, 9 Beta and 9.1.

FILE FORMAT/ENCODING FORMAT	SAMPLING RATE	BIT RATES	NOTES
WMA - Constant Bit Rate (CBR)	At sampling rates of 22.05 KHz playback supported only at specified bit rates	32 kbps (stereo) playback supported	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported

	At sampling rates of 32 KHz playback supported only at specified bit rates	32, 36, 40, 44 and 48 kbps (stereo) playback supported	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
	At sampling rates of 44.1 KHz playback supported only at specified bit rates	32 and 48 kbps (mono); 32, 48, 64, 80, 96, 128, 160 & 192 kbps (stereo) playback supported	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported
	At sampling rates of 48 KHz playback supported only at specified bit rates	64, 96, 128, 160 & 192 kbps (stereo) playback supported	'Lossless', 'Professional' or 'Voice' format files created in Windows Media Player Version 9.0 cannot be supported. DRM (Digital Rights Management) protected files cannot be supported

CD Data Disk AAC Files (only if file extension is '.aac' or '.m4a'): Playback of AAC audio files encoded in Variable Bit Rate (VBR) format is supported at bit rates between 8-320 kbps. For Constant Bit Rate (CBR) files, see table below for compatible sampling rates and bit rates.

FILE FORMAT/ENCODING FORMAT	SAMPLING RATE	BIT RATES	NOTES
AAC - Constant Bit Rate (CBR)	At sampling rates of between 8-24 KHz playback supported at specified bit rates	32-40 kbps; 48-80 kbps; 96-128* kbps playback supported	*112 & 128 kbps bit rate playback not supported for audio files with a sampling rate of 8 KHz
	At sampling rates of between 16-32 KHz playback supported at specified bit rates	160-256* kbps playback supported	*224 & 256 kbps bit rate playback not supported for audio files with a sampling rate of 16 KHz

	At sampling rates of between 32-48 KHz playback supported at specified bit rates	48-80 kbps; 96-128 kbps; 160-256 kbps; 320 kbps playback supported	-
--	--	--	---

PORTABLE AUDIO INTERFACE PANEL/USB POWER SUPPLY

If DTC B1252-19 (USB Port General Electrical Failures - Circuit current above threshold) has been logged, the IAM has detected a current draw from the Portable Audio Interface Panel in excess of 1.7 Amps. In these circumstances, the power supply to the Portable Audio Interface Panel will be cut for an ignition cycle. To reset the system and restore power, the vehicle needs to be locked (with ignition off) and armed for at least 5 minutes.

The USB port on the Portable Audio Interface Panel is able to supply current to a maximum of 500mA. It should be noted that any portable devices connected via the USB port that required more power may not charge or power up correctly and that this may affect the operation of this device with the infotainment system.

APPLE DEVICE COMPATIBILITY

The following table lists some Apple devices and their compatibility with the information and entertainment system using a USB cable.

FULLY SUPPORTED	PARTIALLY SUPPORTED	NOT SUPPORTED
■ iPod® Classic - 6th/7th generation ■ iPod® Nano - 3rd/4th/5th/6th generation ■ iPod® Touch - 2nd/3rd/4th generation ■ iPhone™ 3/3S ■ iPhone™ 4/4S	■ iPod® Classic - 4th/5th generation ■ iPod® Nano - 1st/2nd generation ■ iPod® Touch - 1st generation ■ iPhone™	■ iPod® Classic - 1st/2nd/3rd generation ■ iPod® Shuffle - 1st/2nd/3rd/4th generation

- iPad™ - 1st generation
(with iOS 4.0 or later)

GEN 2.1 AUDIO AMPLIFIER DIAGNOSTICS

The symptom chart below should be used when diagnosing audio amplifier module faults

REPORTED SYMPTOM	SYMPTOM DESCRIPTION	POTENTIAL CAUSES (FOR GUIDANCE ONLY)	RECOMMENDED ACTION
▪ Complete loss of audio from infotainment system	▪ No audio functions or no sound can be heard when audio function is selected	▪ Audio amplifier module not correctly installed ▪ Circuit connectors to audio amplifier module disconnected or not correctly secured	▪ 1. Check TOPIx for any audio system, audio amplifier module and MOST related SSM/TSBs and perform the specified rectifications as required. Retest to confirm if the issue has been resolved. If the fault is still present, go to next step
▪ Greyed out AV button - No response to audio function commands	▪ Audio functions are not available	▪ MOST ring break ▪ Audio amplifier module fuse failure ▪ Data communication error between audio amplifier module and integrated audio module ▪ Power feed not present or power/ground circuit fault ▪ Audio amplifier	▪ 2. Connect the manufacturer approved diagnostic system, check for DTCs and any recommended software updates. If DTCs are present or if software updates are required, perform the necessary remedial actions as specified by the SDD tool. After performing the necessary actions, confirm if the customer fault is still present. If no DTCs are logged and/or no software updates are required, go to the next step ▪ 3. Using the manufacturer approved diagnostic system, run the audio amplifier module On Demand Self-Test (ODST) routine and check for any DTCs that are logged. If DTCs are present, perform the necessary remedial actions as specified by the SDD tool. If the customer fault is still

			module internal failure	present after performing ODSST routine and rectifying DTCs, go to the next step ■ 4. Using the manufacturer approved diagnostic system, perform a vehicle reset by selecting the "Special Application – vehicle reset" routine. Check for DTCs and confirm if the fault is still present. If issue is still present, go to the next step ■ 5. Gain access to the audio amplifier module and check the security and integrity of the cable connections to the audio amplifier module with reference to the circuit diagrams. Rectify as required ■ 6. With the infotainment system switched on, disconnect the MOST connector from the audio amplifier module and check that a red light is present at the connector end. This will be steady for a second or two and then start flashing. If no light is present, there is a problem in the MOST ring before the audio amplifier module. Other modules on the MOST ring must be investigated further ■ 7. Perform an audio amplifier module hard reset by disconnecting the power cable to the audio amplifier module for at least 3 minutes. Reconnect the power cable and retest. If the fault is still present, go to the next step ■ 8. Replace the audio amplifier module and configure as per the instructions outlined on the SDD tool. Confirm if the fault is still present. If issue is still present go to next step ■ 9. If the fault is still present following replacement of the audio amplifier module,	
--	--	--	------------------------------------	---	--

			contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual
▪ Loss of audio from one or more channels	▪ Partial loss of sound	▪ Audio amplifier module not correctly installed	▪ 1. Verify the reported fault by performing the following checks: ▪ Reset fade and balance settings to the centre of the vehicle ▪ Confirm if the issue reported on the customer's vehicle occurs just after engine start up or after the audio system has been running for 10 minutes or more ▪ Confirm if the fault is present on all audio sources (CD, Radio, DAB Radio/SDARS, USB, Aux) – if the fault is not present on all sources, investigate affected audio source e.g. if issue is only seen on DAB Radio, check DAB module is functioning correctly and the software is up to date ▪ Confirm if the issue is seen on stereo and surround sound settings (if applicable) ▪ 2. Check TOPIx for any audio system, audio amplifier module and MOST related SSM/TSBs and perform the specified rectifications as required. Retest to confirm if the issue has been resolved. If the fault is still present, go to next step ▪ 3. Connect the manufacturer approved diagnostic system, check for DTCs and any recommended software updates. If DTCs are present or if software updates are required, perform the
▪ Audio output crackle or distortion	▪ No significant extraneous noise, but the audio reproduction is not as expected e.g. it is not clean sounding	▪ Circuit connectors to audio amplifier module disconnected or not correctly secured ▪ Incorrect EQ file installed on the audio amplifier module ▪ Water ingress ▪ Speaker fault ▪ Data communication error between audio amplifier module and integrated audio module ▪ Incorrect software installed on the audio amplifier module ▪ Intermittent ground circuit connection fault ▪ Audio amplifier module internal failure	

necessary remedial actions as specified by the SDD tool. After performing the necessary actions, confirm if the customer fault is still present. If no DTCs are logged and/or no software updates are required, go to the next step

- 4. Using the manufacturer approved diagnostic system, run the audio amplifier module On Demand Self-Test (ODST) routine and check for any DTCs that are logged. If DTCs are present, perform the necessary remedial actions as specified by the SDD tool. If the customer fault is still present after performing ODST routine and rectifying DTCs, go to the next step
- 5. Using the manufacturer approved diagnostic system, perform a vehicle reset by selecting the "Special Application – vehicle reset" routine. Check for DTCs and confirm if the fault is still present. If issue is still present, go to the next step
- 6. Gain access to the audio amplifier module and check the security and integrity of the cable connections to the audio amplifier module with reference to the circuit diagrams. Rectify as required
- 7. With the infotainment system switched on, disconnect the MOST connector from the audio amplifier module and check that a red light is present at the connector end. This will be steady for a second or two and then start flashing. If no light is present, there is a problem in the MOST ring before the audio amplifier module. Other modules on the MOST ring must be

			investigated further 8. Perform an audio amplifier module hard reset by disconnecting the power cable to the audio amplifier module for at least 3 minutes. Reconnect the power cable and retest. If the fault is still present, go to the next step 9. Replace the audio amplifier module and configure as per the instructions outlined on the SDD tool. Confirm if the fault is still present. If issue is still present go to next step 10. If the fault is still present following replacement of the audio amplifier module, contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual
--	--	--	---

MICROPHONE DIAGNOSTICS

The symptom chart below should be used when diagnosing microphone faults

REPORTED SYMPTOM	SYMPTOM DESCRIPTION	POTENTIAL CAUSES (FOR GUIDANCE ONLY)	RECOMMENDED ACTION
Voice command is not working	When pressing the voice command button on the steering wheel, the system is not responding	- Issue with steering wheel switches - No microphone(s) installed - Microphone not connected correctly - Faulty	1. Check and confirm customer's vehicle does have voice command capability installed on the vehicle and that the voice command function is enabled 2. The microphone is shared between the Bluetooth telephone function and the voice command function. Check that the voice command function works by giving a voice command, for example "Telephone Help" or

		microphone	"Navigation Help". Check that Bluetooth telephone calls can be made. Ensure that the following criteria are satisfied before any telephone call is attempted:
Bluetooth telephone – 3rd party hears interference	When customer uses the telephone via a Bluetooth connection, the call receiver hears interference over the caller's audio signal	- The phone/device is incompatible with JLR infotainment system - Poor (sub-optimal) placement of the phone/device within the vehicle - Poor mobile phone network reception - High mobile phone network demand - Damaged microphone harness - Microphone unclipped from housing - No microphone(s) installed - Microphone not connected correctly - Faulty microphone	- The telephone handset and associated level of software is included on the JLR approved list - The telephone/device battery is fully charged and in a serviceable condition - There is a reliable telephone network reception signal of suitable strength - The telephone/device is placed within the vehicle cabin area - The telephone/device is connected to the vehicle via Bluetooth
Bluetooth telephone – 3rd party cannot hear caller at all	When customer uses the telephone via a Bluetooth connection, the call receiver cannot hear the caller		If the voice command function works and the phone call receiver can hear the caller adequately when calls are made from the vehicle, this suggests that the fault is not with the microphone 3. If the phone call was unsuccessful and the voice command is inoperative, first check if there are any SSMs/TSBs for voice command functions, microphones or Bluetooth and perform the specified rectifications as required. Retest to confirm if the issue has been resolved. If the fault is still present, go to next step 4. Connect the manufacturer approved diagnostic system, check for related DTCs and any recommended software updates. If DTCs are present or if software updates are required, perform the necessary remedial actions as specified by the SDD tool. After performing the necessary actions, confirm if the customer fault is still present. If

			no DTCs are logged and/or no software updates are required, go to the next step ▪ 5. Gain access to the microphone to check the following: ▪ Check the microphone is fitted to driver's side of the vehicle and rectify as required ▪ Check the harness assembly is not trapped and rectify as required ▪ Check the harness assembly connections are fully inserted and secured to the microphone and rectify as required ▪ Check for any harness damage and repair as required ▪ 6. Check the microphone for the following and replace the component if any damage is evident: ▪ Check the microphone connector is free from damage and corrosion ▪ Check for visual damage to the microphone e.g. wiring or casing ▪ 7. If the fault is still present following replacement of the microphone, contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual
--	--	--	---

WHITEFIRE WIRELESS HEADPHONES – FURTHER DIAGNOSTICS

For diagnosis of issues with wireless headphones operation, see symptom chart below:

NOTES:

- If a headphone concern is identified on a vehicle based in the United Kingdom market, please submit an Electronic Product Quality Report (ePQR) prior to carrying out diagnosis as JLR engineering would like to visit the vehicle to assist with diagnosis. If the customer is not prepared to wait for an extended period of time to allow for an engineering visit, then please continue with diagnosis and rectification
- There are up to three audio channels available for wireless headphones, depending on the vehicle specification. 1 - wireless headphones (dual view touch screen output only); 2 - wireless headphones (rear seat entertainment output only); 3 - wireless headphones (dual view touch screen and rear seat entertainment output)

SYMPTOM	POSSIBLE CAUSES	ACTION
Wireless headphones status LED is permanently off	■ Wireless headphones have not been switched on	■ Press the power switch on the right side of the wireless headphones
	■ No batteries are installed in the wireless headphones	■ Remove battery cover on the left side of the wireless headphones and insert 2 x new AAA batteries
	■ Batteries are inserted with incorrect polarity	■ Check the batteries are installed as per the diagram in the battery compartment
	■ Batteries are depleted	■ Remove battery cover on the left side of the wireless headphones and replace batteries with 2 x new AAA batteries

	Wireless headphones have timed out and automatically switched off	Press the power switch on the right side of the wireless headphones
If the headphones status LED remains inactive following completion of all remedial actions detailed above then replace the headphones		
Wireless headphones status LED flashing (red or amber) and no audio from headphone. This indicates the wireless headphones are switched on ready to receive a signal from the vehicle overhead console emitter/infotainment system	Overhead console emitter is not active (an array of red LEDs can be seen when active)	- Ensure the vehicle infotainment system is on and the dual view and/or rear seat entertainment features are active - Select the same audio source to cabin speakers to confirm audio output is operational - If the vehicle overhead console emitter LEDs are not activated following completion of the remedial actions detailed above then continue with diagnosis of the audio system with reference to TOPIx and via Symptom Driven Diagnostics (SDD) which may be accessed using the manufacturer-approved diagnostic tool
	Wireless headphones are not in direct line of sight with the overhead console emitter	Ensure there is a direct line of sight between the overhead console emitter and the wireless headphones
	Wireless headphones are not being used inside the vehicle	The wireless headphones are designed to be used whilst seated in the vehicle only
If, following completion of all remedial actions detailed above, there is no audio output from the headphones when the overhead console emitter LEDs are active and there is a direct line of sight between the overhead console emitter and the wireless headphones, continue to next step		

Wireless headphones status LED constantly on (red or amber) and no audio output from wireless headphones. This indicates wireless headphones are switched on and receiving a signal from the entertainment system via the overhead console emitter	Audio source selected is not playing (CD paused or radio station not tuned in)	- Ensure the active audio source is playing - Select the same audio source to cabin speakers to confirm audio output is operational
	Volume control is set to minimum	Rotate the volume control located on the right side of the wireless headphones in a clockwise direction to increase volume
	Wireless headphones are set to an inactive zone channel. (There are up to 3 channels depending on vehicle specification - see note below)	Press the channel select switch on the right side of the wireless headphones. A beep will be heard from the wireless headphones when the channel is changed. Repeat if necessary
	Wireless headphones audio is inhibited in dual view touch screen or rear seat entertainment control menu	Enable the wireless headphones audio via the dual view touch screen or rear seat entertainment control menu
If there is no audio output from the headphones (or audio output from only one earpiece) following completion of all remedial actions detailed above then replace the headphones		

PINPOINT TESTS

NOTE:

Soft key confirmation tones will not be audible if this preference is set to off (Set-up / System / Button feedback)

NOTES:

- Operating the steering wheel mode switch briefly will cycle through the audio/video sources as follows: Radio - DAB/SDARS - My Music - My Video
- Operating the steering wheel mode switch for >2 seconds will cycle through the minor sources as follows: (Radio) FM1 - FM2 - FM3 - AM1 - AM2, and (My Music) DVD/CD - iPod® - BT - Stored CDs - USB - AUX
- Depending on vehicle specification, various audio/video sources may be installed and are contained in the control modules as follows:
 - Integrated Audio Module (IAM) - including radio, CD, iPod®, USB, my music, my video, phone and navigation
 - Digital Radio Control Module (DRCM)
 - Satellite Radio Control Module (SRCM)
 - Television Control Module (TVCM)
 - Rear Seat Entertainment Control Module (RSECM)

PINPOINT TEST A : SOURCE TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: SOURCE TEST 1	
	1 Operate the available soft keys
	Was there an audible confirmation tone to indicate that the soft key

	input was detected? Yes Audio amplifier module and MOST network functioning. GO to A2. No Audio amplifier module fault or MOST ring break. GO to A2.
--	---

A2: SOURCE TEST 2

	1 Cycle through the audio/video sources by operating the steering wheel mode switch
	Did the audio/video soft key return to normal and/or the selected source function normally? Yes MOST network functioning. GO to A3. No Possible MOST ring break. GO to A3.

A3: SOURCE TEST 3

	1 Operate the Navigation soft key (or switch)
	Did the navigation system start up and display a map? Yes GO to A4. No GO to A4.

A4: SOURCE TEST 4

	1 Operate the Phone soft key (or switch)
	Is the phone menu displayed? Yes GO to Pinpoint Test B. No GO to Pinpoint Test B.

NOTE:

Depending on vehicle specification, various audio/video sources may be installed and are contained in the control modules as follows:

- Integrated Audio Module (IAM) - including radio, CD, iPod®, USB, my music, my video, phone and navigation
- Digital Radio Control Module (DRCM)
- Satellite Radio Control Module (SRCM)
- Television Control Module (TVCM)
- Rear Seat Entertainment Control Module (RSECM)

PINPOINT TEST B : FUSE PULL TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: FUSE PULL TEST 1	
	1 Remove the fuse from the missing audio/video source control module circuit
	2 Inspect the fuse
	Has the fuse blown? Yes Refer to the electrical circuit diagrams and test the circuit for short circuit to ground. Repair as necessary. Install a new fuse. GO to B2. No Wait at least 30 seconds and re-install the fuse. GO to B2.
B2: FUSE PULL TEST 2	
	1 Set the ignition to off
	2 Set the ignition to on
	3 Check the operation of the touch screen and all audio/video sources
	Has full audio/video functionality been restored? Yes If the missing source was part of the integrated audio module, using the manufacturer approved diagnostic system, re-configure the integrated audio module with the latest level software. Tests complete No

	GO to B3.
B3: FUSE PULL TEST 3	
	1 Refer to the electrical circuit diagrams and identify the next control module connected to the MOST network
	Is there another control module that has not been reset? Yes GO to B4. No MOST ring break present. REFER to: Communications Network (418-00 Module Communications Network, Diagnosis and Testing).
B4: FUSE PULL TEST 4	
	1 Remove the fuse from the next control module circuit
	2 Inspect the fuse
	Has the fuse blown? Yes Refer to the electrical circuit diagrams and test the circuit for short circuit to ground. Repair as necessary. Install a new fuse. GO to B2. No Wait at least 30 seconds and re-install the fuse. GO to B2.

NOTE:

A suitable stylus (for example the stylus supplied with a Panasonic CF-19) will be required for this procedure.

PINPOINT TEST C : TOUCH SCREEN CALIBRATION	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: TOUCH SCREEN CALIBRATION	
	1 Operate the Valet soft key continuously until the engineering mode screen is displayed (approximately 20 seconds)
	2 Scroll down and select Touch Calibration
	3 Select OK
	4 Tap the touch screen to proceed
	5 Using a suitable stylus, tap the touch screen at the points indicated

	until a pass/fail result is displayed
	Was the touch screen calibration successful? Yes Calibration complete No Calibration failed. GO to C1.

DIAGNOSTICS FOR INCONTROL™ APPLICATIONS (APPLIES TO VEHICLES FITTED WITH TELEPHONE INTERFACE MODULE)

The following diagnostics should be used when customers are reporting issues with the access or operation of infotainment InControl™ applications. Check for related DTCs logged in the Telephone Interface Module (TIM) and, where possible, rectify these prior to proceeding with the diagnostic procedures described below. If no related DTCs are logged, proceed with the diagnostic procedures described below.

ACTIVATING THE INCONTROL™ APPS SYSTEM DIAGNOSTICS

To access the InControl™ Apps system integral diagnostics follow the steps detailed below:

1. Activate the InControl™ Apps system by pressing the InControl™ Apps key displayed on the touch screen
2. Press the 'Settings' key (found in the bottom left corner of the InControl™ Apps connection screen)
3. Press the blank space in the bottom left of the settings menu screen for at least 12 seconds, until the display changes to the diagnostics PIN entry screen
4. Input PIN 753 to enter the diagnostics screens
5. Press the 'Next' key on the diagnostic screen to toggle between the

SOFTWARE UPDATES FOR THE INCONTROL™ APPS SYSTEM

NOTES:

- The software update cannot be performed whilst the vehicle is in motion, the software update must be a valid InControl™ Apps version and the USB flash drive must not be removed until prompted to do so
- To facilitate a successful software update, it is recommended that the functionality of the USB stick should be confirmed before attempting the update. In addition, it is recommended that any software other than the intended update (including previous versions of InControl™ Apps updates) should be removed from the USB stick before attempting the update

Periodic software updates for the InControl™ Apps system may be uploaded to the telephone interface module via a USB stick connected to the appropriate portable audio interface panel port. Software updates may be downloaded from www.landrover.com or www.jaguar.com. To perform the software update process follow the steps detailed below:

1. Activate the InControl™ Apps system by pressing the InControl™ Apps key displayed on the touch screen
2. Press the 'Settings' key (found in the bottom left corner of the InControl™ Apps connection screen)
3. Press the 'Software' key on the settings menu screen, then press the 'Update' key
4. Connect a USB stick loaded with the software update to the

appropriate portable audio interface panel port. Once the system has scanned the USB stick and verified the software update, a 'Next' key will be activated on the touch screen

5. Press the 'Next' key. At the next screen, confirm that the software update is the correct version, then press the 'Next' key to initiate the software update. A status bar will display the progress of the update
6. Once the update is complete, remove the USB stick and press the 'Finish' key

SYMPTOM CHART

SYMPTOM	ACTION
The "InControl™ Apps Not Available" pop-up is displayed when the InControl™ Apps key is pressed from the home screen displayed on the touch screen	▪ GO to Pinpoint Test D.
There is only a black screen shown when attempting to access InControl™ Apps	▪ GO to Pinpoint Test E.
The customer reports that when InControl™ Apps key is pressed on the touch screen the 'Not Connected' screen is seen and their phone/device does not connect to InControl™ Apps	▪ GO to Pinpoint Test F.
A connection to the customer's device is established as soon as it is connected to the portable audio interface panel USB port, without the InControl™ Apps key being pressed on the touch screen	▪ GO to Pinpoint Test G.
'Auto Connect' does not work when a phone/device is connected to the appropriate portable audio interface panel port	▪ GO to Pinpoint Test H.
The customer is unable to make phone calls through InControl™ Apps	▪ GO to Pinpoint Test I.
Nothing is heard when audio applications are used in InControl™ Apps	▪ GO to

	Pinpoint Test J.
The 'Next' button remains greyed out in the software update screen when a USB stick is connected	▪ GO to Pinpoint Test K.
The 'Vehicle in motion' pop-up is displayed when attempting a software update	▪ GO to Pinpoint Test L.
The 'Vehicle in motion' pop-up is displayed	▪ GO to Pinpoint Test M.
An app that is speed inhibited can be opened while the vehicle is in motion	▪ GO to Pinpoint Test N.
The GPS position in the navigation apps is incorrect	▪ GO to Pinpoint Test O.

PINPOINT TESTS FOR INCONTROL™ APPS ISSUES

PINPOINT TEST D : THE "INCONTROL™ APPS NOT AVAILABLE" POP-UP IS DISPLAYED WHEN THE INCONTROL™ APPS KEY IS PRESSED FROM THE HOME SCREEN DISPLAYED ON THE TOUCH SCREEN

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: CHECK TIM FUSES	
	1 Check the fuses for the telephone interface module
	Are any of the fuses blown? Yes Replace fuse(s) as required No GO to D2.
D2: CHECK THE TOUCH SCREEN SOFTWARE VERSION	

	1 Check the touch screen has the latest software version installed
	Is the latest version of touch screen software installed? Yes GO to D3. No Update the touch screen software as required
D3: CHECK THE TIM POWER SUPPLY VOLTAGE	
	1 Using the manufacturer approved diagnostic tool, check the telephone interface module power supply voltage
	Is the telephone interface module power supply voltage within the correct range (between 11-15 volts)? Yes GO to D4. No Check the telephone interface module power supply circuits for circuit faults and rectify as required. Refer to the relevant section of the workshop manual and test the battery and charging system
D4: CHECK TIM CONNECTOR 1	
	1 Disconnect connector 1 from the telephone interface module and check for any obvious damage (i.e. damaged casing, broken wires, poorly seated or damaged pins)
	Is there any damage to the connector and/or pins? Yes Replace/repair the connector as required. Then, GO to D5. No GO to D5.
D5: RE-COUPLE CONNECTOR AND RECHECK	
	1 Reconnect connector 1 to the telephone interface module
	Is the fault still apparent? Yes GO to D6. No The connector was poorly seated. No further action
D6: CHECK SUPPLY VOLTAGE AT CONNECTOR 1	
	1 Disconnect connector 1 from the telephone interface module and measure the voltage between pin 12 (+V) and pin 7 (0V) of the connector
	Is the voltage measured between 11V and 15V? Yes GO to D7. No Check the telephone interface module power supply circuits for circuit

	faults and rectify as required. Refer to the relevant section of the workshop manual and test the battery and charging system
D7: CHECK CAN VOLTAGES	
	1 Using a digital multimeter, check the voltages at pins 5, 6, 9 and 10
	Do the recorded voltages match these values? Pin 5 = 2V; Pin 6 = 3V; Pin 9 = 2V; Pin 10 = 3V Yes Replace the telephone interface module (TIM) No Check the CAN circuits for circuit faults and rectify as required

PINPOINT TEST E : THERE IS ONLY A BLACK SCREEN SHOWN WHEN ATTEMPTING TO ACCESS INCONTROL™ APPS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: CHECK THE LVDS CONNECTORS BETWEEN THE TOUCH SCREEN AND THE TIM (1)	
	1 Remove the blue LVDS connector from the touch screen and check for obvious damage to the connector or lead
	Is there any damage to the LVDS connector/lead? Yes Replace the LVDS lead between the touch screen and the telephone interface module No Reconnect the blue LVDS connector, then retest. If the fault is rectified, the connector was poorly seated and no further action is required. If the fault persists, then GO to E2 .
E2: CHECK THE LVDS CONNECTORS BETWEEN THE TOUCH SCREEN AND THE TIM (2)	
	1 Remove the brown LVDS connector (Connector 5) from the telephone interface module and check for obvious damage to the connector or lead
	Is there any damage to the LVDS connector/lead? Yes Replace the LVDS lead between the touch screen and the telephone interface module No Reconnect the brown LVDS connector, then retest. If the fault is rectified, the connector was poorly seated and no further action is required. If the fault persists, then GO to E3 .
E3: CHECK THE CONTINUITY OF THE LVDS LEAD BETWEEN THE TOUCH SCREEN AND THE TIM	

	1 Using a digital multimeter, check the continuity of the wiring between the corresponding pins of the LVDS connectors at either end of the LVDS lead between the touch screen and the telephone interface module
	Were any continuity faults found? Yes Replace the LVDS lead between the touch screen and the telephone interface module No Replace the touch screen. Reconnect the LVDS connector, then retest. If fault persists, replace the telephone interface module

PINPOINT TEST F : THE CUSTOMER REPORTS THAT WHEN INCONTROL™ APPS KEY IS PRESSED ON THE TOUCH SCREEN THE 'NOT CONNECTED' SCREEN IS SEEN AND THEIR PHONE/DEVICE DOES NOT CONNECT TO INCONTROL™ APPS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: WHICH SCREEN/MESSAGE IS DISPLAYED WHEN THE PHONE IS CONNECTED TO THE SYSTEM?	
	1 Connect the phone to the system via the portable audio interface panel
	Does the touch screen show the 'Not Connected' screen (this screen shows the text "connect compatible device to an identified port in the cubby box")? Yes GO to F2. No GO to F5.
F2: USB DEVICE COMPATIBILITY CHECK	
	1 Check for the 'USB Device Not Supported' message
	When the phone is first connected, is the 'USB device is not supported' message displayed? Yes The phone/device is not supported by InControl™ system. Refer to the InControl™ system supported device list (available online - http://jaguar50.nextgen-technology.net/ or http://landrover50.nextgen-technology.net/) and inform customer accordingly No GO to F3.
F3: APPLE DEVICE COMPATIBILITY CHECK	
	1 Check for the 'Device not supported' message
	When the phone is first connected, is the 'Device not supported. For

	iPod functionality, please connect device to identified port in the cubby box and go to Audio/Video' message displayed? Yes The Apple phone/device is not supported by InControl™ system. Refer to the InControl™ system supported device list and inform customer accordingly No GO to F4.
--	---

F4: CHECK INSTALLED DEVICE APPLICATIONS

	1 Check for the 'Ensure device has the InControl™ Apps application installed' message
	When the phone is first connected, is the 'Ensure device has the InControl™ Apps application installed and is in an unlocked state' message displayed? Yes Ensure that the customer's device has the correct InControl™ Apps app installed and that the device is in an unlocked state when connected to the InControl™ system. If the phone has the correct app and is unlocked but this screen is still seen, shut down the InControl™ Apps phone app and try again. If this does not work, turn the phone off and then back on No GO to F5.

F5: FURTHER USB CONNECTION ISSUES - CORRECT USB PORT USED?

	1 Check if the correct USB port is being used for connection of the device to the system
	Is the correct USB port being used for connection of the device to the system? Yes GO to F6. No Inform customer which USB port should be used for the connection of the device to the system

F6: FURTHER CONNECTION ISSUES - CHECK OPERATION WITH DIFFERENT DEVICE/LEAD?

	1 Test system by connecting a different device and lead to the appropriate port in the portable audio interface panel
	Does the system attempt a connection when a different device and/or lead is used? Yes The fault may be with either the USB/Lightning lead or with the customer's phone. Advise customer accordingly No GO to F7.

F7: CHECK CABLE BETWEEN THE PORTABLE AUDIO INTERFACE PANEL AND THE TIM (1)

	1 Remove the connector from the portable audio interface panel and check for obvious damage to the connector or lead
	Is there any obvious damage to the lead that connects the portable audio interface panel to the telephone interface module? Yes Replace the lead as required No Reconnect the lead that connects the portable audio interface panel to the telephone interface module, then retest. If the fault is rectified, the connector was poorly seated in the portable audio interface panel socket and no further action is required. If the fault persists, GO to F8.

F8: CHECK CABLE BETWEEN THE PORTABLE AUDIO INTERFACE PANEL AND THE TIM (2)

	1 Remove Connector 2 from the telephone interface module and check for obvious damage to the connector or lead
	Is there any obvious damage to the lead that connects the portable audio interface panel to the telephone interface module? Yes Replace the lead as required No Reconnect the lead that connects the portable audio interface panel to the telephone interface module, then retest. If the fault is rectified, the connector was poorly seated in the telephone interface module socket and no further action is required. If the fault persists, GO to F9.

F9: CHECK THE CONTINUITY OF THE LEAD BETWEEN THE PORTABLE AUDIO INTERFACE PANEL AND THE TIM

	1 Using a digital multimeter, check the continuity of the wiring between the corresponding pins of the connectors at either end of the lead between the portable audio interface panel and the telephone interface module
	Were any continuity faults found? Yes Replace the lead between the portable audio interface panel and the telephone interface module No Replace the portable audio interface panel. Reconnect the lead between the portable audio interface panel and the telephone interface module, then retest. If fault persists, replace the telephone interface module

PINPOINT TEST G : A CONNECTION TO THE CUSTOMER'S DEVICE IS ESTABLISHED AS SOON AS IT IS CONNECTED TO THE PORTABLE AUDIO INTERFACE PANEL USB PORT, WITHOUT THE INCONTROL™ APPS KEY BEING PRESSED ON THE TOUCH SCREEN

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
-----------------	-------------------------

G1: CHECK THE STATUS OF THE 'AUTO CONNECT' FUNCTION

	1 Navigate to the InControl™ Apps Display screen via the Settings key
	Is the 'Auto Connect' function in the Display screen switched to off? Yes Turn the 'Auto Connect' function off, turn off the ignition, step out of the vehicle and lock the doors. Leave the vehicle for at least 10 minutes so the vehicle CAN bus goes to sleep and a full shut down occurs. Enter and restart the vehicle then, GO to G2. No If the 'Auto Connect' function in the Display screen switched to on, then a connection to the customer's device is established as soon as it is connected to the portable audio interface panel under normal operation

G2: RECHECK FOR FAULT

	1 Recheck if a connection to the customer's device is established as soon as it is connected to the portable audio interface panel USB port, without the InControl™ Apps key being pressed on the touch screen
	Is the fault still apparent? I.e: a connection to the customer's device is established automatically, despite the 'Auto Connect' function in the Display screen being switched to off Yes Replace the telephone interface module No The customer may not have turned the 'Auto Connect' function off or the vehicle may not have performed a full shut down previously. Show the customer how to use the 'Auto Connect' feature. If the customer reports that the problems persists, investigate faults that may prevent the vehicle's CAN bus from shutting down correctly

PINPOINT TEST H : 'AUTO CONNECT' DOES NOT WORK WHEN A PHONE/DEVICE IS CONNECTED TO THE APPROPRIATE PORTABLE AUDIO INTERFACE PANEL PORT

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: CHECK THE STATUS OF THE 'AUTO CONNECT' FUNCTION	
	1 Navigate to the InControl™ Apps Display screen via the Settings key
	Is the 'Auto Connect' function in the Display screen switched to on? Yes Update the touch screen with the latest software and then, GO to H4. No Turn the 'Auto Connect' function on, turn off the ignition, step out of the vehicle and lock the doors. Leave the vehicle for 10 minutes so the vehicle CAN bus goes to sleep and a full shut down occurs. Enter and restart the vehicle, then GO to H2.

H2: RECHECK FOR FAULT

	1 Recheck if a connection to the customer's device is not established as soon as it is connected to the portable audio interface panel USB port
	Is the fault still apparent? I.e: a connection to the customer's device is not established automatically, despite the 'Auto Connect' function in the Display screen being switched to on Yes GO to H3. No The customer may not have turned on the 'Auto Connect' function. Show the customer how to operate the 'Auto Connect' feature

H3: RECHECK THE STATUS OF THE 'AUTO CONNECT' FUNCTION

	1 Navigate to the InControl™ Apps Display screen via the Settings key
	Is the 'Auto Connect' function in the Display screen still switched to on? Yes Update the touch screen with the latest software and then, GO to H4. No Replace the telephone interface module

H4: RECHECK FOR FAULT

	1 Recheck if a connection to the customer's device is not established as soon as it is connected to the portable audio interface panel USB port
	Is the fault still apparent? I.e: a connection to the customer's device is not established automatically, despite the 'Auto Connect' function in the Display screen being switched to on Yes Replace the touch screen No The software update has cleared the fault. No further action required

PINPOINT TEST I : THE CUSTOMER IS UNABLE TO MAKE PHONE CALLS THROUGH INCONTROL™ APPS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: CHECK BLUETOOTH SETTINGS	
	1 Check if the phone has Bluetooth switched on
	2 Check if the phone is connected (paired) via Bluetooth to the infotainment system
	Is the phone Bluetooth paired and connected to the infotainment system? Yes GO to I2.

	No The customer's phone must be Bluetooth paired before a call can be placed. Activate the device's Bluetooth function and connect the device to the infotainment system. Inform the customer how to do this operation for future reference
--	--

I2: CHECK FOR ERROR MESSAGE SCREEN

	1 With the phone connected (paired) via Bluetooth to the infotainment system, check for any error messages displayed on the touch screen
	Is the error message "To make or receive phone calls connect your phone to Bluetooth Handsfree" shown? Yes The device is not set as the Bluetooth phone source. Configure the infotainment system accordingly. Inform the customer how to do this operation for future reference No GO to I3.

I3: CHECK IF THE CONNECTED DEVICE IS CAPABLE OF MAKING CALLS

	1 Check the connected device is capable of making calls
	Is the connected device capable of making calls? (e.g. check that the phone is not call-barred or that it is not an iPod®) Yes Check if the phone can make calls when operating independently of the InControl™ Apps system(i.e. using the infotainment system hands free function). If calls are still not possible, check the operation of the infotainment system and rectify as required No Disable the phone's call-barring function or connect a suitable device. Inform the customer how to do this operation for future reference

PINPOINT TEST J : NOTHING IS HEARD WHEN AUDIO APPLICATIONS ARE USED IN INCONTROL™ APPS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
J1: CHECK THE INFOTAINMENT AUDIO VOLUME SETTINGS	
	1 Check the audio volume level of the infotainment system
	Is the infotainment system volume at an audible level? Yes GO to J2. No Adjust the volume setting and inform the customer how to do this operation for future reference
J2: CHECK FOR AUDIO FROM OTHER SOURCES	

	1 Check that audio output is possible when other infotainment sources (i.e. Radio/CD/DVD) are selected
	Is there audio available from the rest of the infotainment system? Yes GO to J3. No Check the operation of the infotainment system and rectify as required

J3: CHECK BLUETOOTH SETTINGS

	1 Check if the phone has Bluetooth switched on
	2 Check if the phone is connected (paired) via Bluetooth to the infotainment system
	Is the phone Bluetooth paired and connected to the infotainment system? Yes GO to J4. No The customer's phone must be Bluetooth paired before audio output is possible. Activate the device's Bluetooth function and connect the device to the infotainment system. Inform the customer how to do this operation for future reference

J4: CHECK THE INCONTROL™ APPS SETTINGS

	1 Check that the 'View and listen' function is selected as opposed to the 'View' only setting
	Are the InControl™ Apps settings set to 'View' and not 'View and listen'? Yes Select the 'View and listen' function. Inform the customer how to do this operation for future reference No GO to J5.

J5: CHECK THE INFOTAINMENT BLUETOOTH AUDIO SOURCE SETTINGS

	1 Check the infotainment Bluetooth audio source settings
	Is the device connected to the appropriate input port with InControl™ Apps selected as the Bluetooth audio source? (It is possible to change the audio source after initiating the InControl™ Apps system) Yes No further action No Change the audio source accordingly. Inform the customer how to do this operation for future reference

PINPOINT TEST K : THE 'NEXT' BUTTON REMAINS GREYED OUT IN THE SOFTWARE UPDATE SCREEN WHEN A USB STICK IS CONNECTED

TEST
CONDITIONS

DETAILS/RESULTS/ACTIONS

K1: CHECK THE CAR CONFIGURATION PARAMETERS

	1 Using the touch screen controls, navigate to the Diagnostics (1) menu within InControl™ Apps and select the Car Configuration Parameters screen
	Do the details shown on the Car Configuration Parameters screen match the actual vehicle? Yes GO to K2. No Reflash the telephone interface module unit with the correct car configuration parameters

K2: CHECK USB STICK CONTENT AND CONNECTION

	1 Check that the USB stick is connected to the correct input port
	2 Check that there is a valid versions of the InControl™ Apps software update loaded on the USB stick
	Is a USB stick loaded with a valid InControl™ Apps software update connected to the correct input port? Yes GO to K3. No Ensure that the USB stick is connected to the correct port and that there is a valid InControl™ Apps software update loaded onto the USB stick. Inform the customer how to do these operations for future reference

K3: CHECK USB STICK FUNCTIONALITY

	1 Check that the USB stick is functioning correctly (try installing the InControl™ Apps software update from another USB device)
	Can alternative USB sticks be used to successfully update the InControl™ Apps software? Yes The USB stick may be faulty. Advise the customer to replace it with a supported/functional USB stick No Check if the InControl™ Apps system attempts to connect to other devices connected to the USB port. If no connection is attempted when a device is connected to the appropriate input port then, GO to Pinpoint Test F.

PINPOINT TEST L : THE 'VEHICLE IN MOTION' POP-UP IS DISPLAYED WHEN
ATTEMPTING A SOFTWARE UPDATE

TEST
CONDITIONS

DETAILS/RESULTS/ACTIONS

L1: CHECK VEHICLE SPEED STATUS DURING THE SOFTWARE UPDATE

	1 Ascertain if the car was in motion during the software update
	Was the car in motion during the software update? Yes Software updates can only be performed when the vehicle is stationary. Advise the customer accordingly No Check if there are any other vehicle faults or DTCs that would infer that the vehicle in motion status is unknown (If no speed signal is received, the InControl™ Apps system will assume the vehicle is in motion and it will be impossible to open speed inhibited apps even if the vehicle is actually stationary). Rectify as required

PINPOINT TEST M : THE 'VEHICLE IN MOTION' POP-UP IS DISPLAYED

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
M1: CHECK VEHICLE SPEED STATUS	
	1 Ascertain if the car was in motion when the pop-up was displayed?
	Was the car was in motion when the pop-up was displayed? Yes GO to M3. No GO to M2.
M2: CHECK THE CAR CONFIGURATION PARAMETERS	
	1 Using the touch screen controls, navigate to the Diagnostics (1) menu within InControl™ Apps and select the Car Configuration Parameters screen
	Do the details shown on the Car Configuration Parameters screen match the actual vehicle? Yes GO to M3. No Reflash the telephone interface module unit with the correct car configuration parameters
M3: CHECK FOR ANY OTHER VEHICLE IN MOTION STATUS FAULTS OR DTCs	
	1 Check for related faults/DTCs that would imply that the vehicle in motion status is unknown
	Are there any other vehicle faults or DTCs that would imply that the vehicle in motion status is unknown? Yes If no speed signal is received, the InControl™ Apps system will assume the vehicle is in motion and it will be impossible to open speed inhibited

	apps even if the vehicle is actually stationary. Rectify as required No GO to M4.
M4: CHECK IF THE APP IN USE WHEN THE POP-UP WAS DISPLAYED WAS A SPEED-INHIBITED APP	
	1 Check the nature of the app being used when the pop-up was triggered
	Was the app being used at the time the pop-up was displayed a speed-inhibited app? Yes Only certain apps can be accessed/displayed when the vehicle is in motion. Advise the customer accordingly No Check that the latest version of the InControl™ Apps app is installed on the phone. Install latest versions as required
PINPOINT TEST N : AN APP THAT IS SPEED-INHIBITED CAN BE OPENED WHILE THE VEHICLE IS IN MOTION	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
N1: CHECK THE FUNCTIONALITY OF OTHER SPEED INHIBITED APPS	
	1 Check if other speed-inhibited apps be opened whilst the vehicle is in motion
	Can other speed-inhibited apps be opened whilst the vehicle is in motion? Yes GO to N3. No GO to N2.
N2: CHECK THE CURRENT SPEED-INHIBITED STATUS OF THE APP	
	1 Check the status of the app
	Has the app concerned been recently removed from the list of speed-inhibited apps? Yes Inform the customer that the app has now been removed from the list of speed-inhibited apps and can be used on the move No GO to N3.
N3: CHECK THE SOFTWARE VERSION OF THE PHONE/DEVICE APP	
	1 Check the version of the InControl™ Apps app software installed on the phone/device

	Does the phone/device have the latest version of the InControl™ Apps app software installed? Yes GO to O1. No Update the phone with the latest version of the InControl™ Apps app software and try opening the app again. If the speed-inhibited app still opens while the vehicle is moving, check whether the same fault occurs when using a different device connected to the system. If the fault cannot be replicated using a different device, it is likely that there is a fault with the original phone/device. Advise the customer accordingly
--	---

PINPOINT TEST O : THE GPS POSITION IN THE NAVIGATION APPS IS INCORRECT	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
O1: CHECK THE FUNCTIONALITY OF THE PHONE/DEVICE OR NAVIGATION APP IN USE	
	1 Check the GPS position data when a different phone/device or a different navigation app is used
	Is the problem still present when using a different phone/device or when using a different navigation app? Yes GO to O2. No If the fault cannot be replicated using a different device or a different navigation app, it is likely that there is a fault with the original phone/device or app. Advise the customer accordingly
O2: CHECK THE NAVIGATION SYSTEM HARDWARE TYPE	
	1 Ascertain the on-board navigation system
	Is the vehicle fitted with a dead reckoning module instead of a full navigation system? Yes GO to O4. No GO to O3.
O3: CHECK THE VEHICLE'S NAVIGATION SYSTEM POSITION DATA	
	1 Check the position data as generated by the on-board navigation system
	Is the vehicle's own navigation system position correct? Yes GO to O4. No Check the operation of the on-board navigation system as this is the

	likely source of the fault. Rectify as required
O4: CHECK FOR CORRECT GPS INFORMATION INPUT TO THE TELEPHONE INTERFACE MODULE	
	1 Navigate to Diagnostic Screen 2 in the InControl™ Apps system
	2 Press the 'GPS Information' button to check the GPS information status
	Does the InControl™ Apps diagnostics screen show GPS information is being received? Yes Reflash the telephone interface module unit with the latest level software No Check the operation of the on-board navigation system as this is the likely source of the fault. Rectify as required

TOUCH SCREEN DIAGNOSTICS

The Touch Screen (TS), also referred to as the Front Control Display Interface Module (FCDIM) or the High Line Display Front (HLDF), may be wrongly replaced when troubleshooting, if issues are incorrectly attributed to it

The TS may be incorrectly diagnosed as the source of the fault and replaced in error. The TS is used only for displaying information to the user which will be generated by other modules in the vehicle. Although the selection is done via the TS, it must wait for a response from the selected module. Therefore, errors in specific functions (see below) should be resolved within the associated control module and should not result in the changing of the TS

DO NOT replace the TS for any of the following issues. Refer to the control module associated with the fault:

- **Audio Quality:** A 'no or poor quality audio' concern should not result in a replaced TS. The amplifier, speakers and/or Integrated Audio Module (IAM) should be checked and thoroughly excluded first
- **Bluetooth Issues:** The IAM controls all bluetooth connectivity. Check the IAM system for correct functionality

- **CD/DVD Playback Issues:** Check the IAM for media integrity and mechanism issues. The eject function is controlled by the IAM although selection is through the TS
- **DAB Radio:** Check the Digital Radio Control Module (DRCM) or associated antennas for correct functionality
- **Greyed Out Buttons:** Lost communications through the Media Oriented Systems Transport (MOST). Check the MOST ring for correct functionality
- **Navigation ROUTING/MAP Issues** (includes all images associated with the navigation system including TRAFFIC UPDATES and VOICE INSTRUCTIONS): The TS only reacts as instructed by the IAM, therefore check the IAM for correct functionality
- **Radio** (includes all issues of clarity, not finding channels, not storing radio channels correctly): Check antenna(s) and/or IAM for correct functionality
- **TV Related Issues:** Check related antenna or IAM for correct functionality

If it is still unclear as to which component may be at fault, please raise a Technical Assistance (TA) request through Dealer Technical Support (DTS).

Note: Warranty claims may be rejected if the above procedure is not followed

SPEAKER DIAGNOSTICS

For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual.

REFER to: [Speakers](#) (415-01 Information and Entertainment System, Diagnosis and Testing).

CAMERA DIAGNOSTICS

For Camera Diagnostics please refer to section 413-13 Diagnosis and Testing - Proximity Camera in the workshop manual.

REFER to: [Proximity Camera](#) (413-13 Parking Aid, Diagnosis and Testing).

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.